# EQUIVALENT CHARACTERISATIONS OF WEAKLY PRE-ADIC RINGS, AND THE COMPLETION OF A WEAKLY PRE-ADIC RING

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The list of equivalent conditions is reproduced with the author's own numbering, including the two additional statements that follow the list rather than being part of the equivalence; the card's obligations keep that separation.

*Editorial note.* The proof contains the author's abbreviation 'ie.' and the sentence 'The implication (4) implies (3) is trivial.'; both lie inside the counted span and are reproduced verbatim.

*Editorial note.* The counted spans carry 5 navigation stubs; no bibliography citation occurs in the spans.

*Editorial note.* The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

*Editorial note.* The cited definition of a weakly pre-adic ring carries a footnote citing [Gabber-Ramero]; since the chapter ships no bibliography, that entry is attached to this card as bibliography.tex, copied character for character from the Stacks Project's own my.bib, so that the citation resolves without any entry being invented.

*Allowed standard prerequisites (editorial).*

- Weakly pre-adic and weakly adic rings, and their ideals. Same chapter, Definition definition-weakly-adic, invoked at proof lines 1676-1678 and in the statement. Fixes the first of the four conditions and the notion of an ideal as in that definition.
- An open ideal in a linearly topologized ring is an ideal of definition only up to containing one, and ideals of definition are open. More on Algebra, Definition more-algebra-definition-topological-ring, invoked at proof lines 1681-1683. Gives that an arbitrary ideal of definition contains a power of the chosen one, which is the step from the third to the fourth formulation.
- Tautness of continuous ring maps is preserved by composing with the completion, and the completion of a linearly topologized ring is complete. Same chapter, Lemmas lemma-completion-taut and lemma-composition-taut, invoked at proof lines 1692-1694, and More on Algebra, Section more-algebra-section-topological-ring, invoked at proof lines 1696-1697. Produce the taut map needed to see that the completion is weakly pre-adic, and its completeness.

*Date*: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 0GX2: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1342–1342. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

*Source reference.* Original lemma 0GX3: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1357–1357. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

*Source reference.* Original definition 07E8: sources/stacks-chow/tags.txt, source lines 9438–9438. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

*Source reference.* Original section 07E7: sources/stacks-chow/tags.txt, source lines 9437–9437. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

**Definition 0.1.** Let $A$ be a linearly topologized ring.

(1) We say $A$ is *weakly pre-adic*[1] if there exists an ideal $I \subset A$ such that the closure of $I^n$ is open for all $n \geq 0$ and these closures form a fundamental system of open ideals.
(2) We say $A$ is *weakly adic* if $A$ is weakly pre-adic and complete[2].

**Lemma 0.2.** *Let $A$ be a linearly topologized ring. The following are equivalent*

(1) *$A$ is weakly pre-adic,*
(2) *there exists a taut continuous ring map $A' \to A$ where $A'$ is a pre-adic topological ring, and*
(3) *$A$ is pre-admissible and there exists an ideal of definition $I$ such that the closure of $I^n$ is open for all $n \geq 1$, and*
(4) *$A$ is pre-admissible and for every ideal of definition $I$ the closure of $I^n$ is open for all $n \geq 1$.*

*The completion of a weakly pre-adic ring is weakly adic. If $A$ is weakly adic, then $A$ is admissible and has a countable fundamental system of open ideals.*

*Proof.* Assume (1). Choose an ideal $I$ such that the closure of $I^n$ is open for all $n$ and such that these closures form a fundamental system of open ideals. Denote $A' = A$ endowed with the $I$-adic topology. Then $A' \to A$ is taut by definition and we see that (2) holds.

Assume (2). Let $I' \subset A'$ be an ideal of definition. Denote $I$ the closure of $I'A$. Tautness of $A' \to A$ means that the closures $I_n$ of $(I')^n A$ are open and form a fundamental system of open ideals. Thus $I = I_1$ is open and the closures of $I^n$ are equal to $I_n$ and hence open and form a fundamental system of open ideals. Thus certainly $I$ is an ideal of definition such that the closure of $I^n$ is open for all $n$. Hence (3) holds.

[1] In [Gabber-Ramero] the authors say $A$ is *c-adic*.
[2] By our conventions this includes separated.

If $I \subset A$ is as in (3), then $I$ is an ideal as in Definition 0.1 and we see that (1) holds. Also, if $I' \subset A$ is any other ideal of definition, then $I'$ is open (see More on Algebra, Definition 07E8) and hence contains $I^n$ for some $n \geq 1$. Thus $(I')^m$ contains $I^{nm}$ for all $m \geq 1$ and we conclude that the closures of $(I')^m$ are open for all $m$. In this way we see that (3) implies (4). The implication (4) $\Rightarrow$ (3) is trivial.

Let $A$ be weakly pre-adic. Choose $A' \to A$ as in (2). By Lemmas 0GX2 and 0GX3 the composition $A' \to A^\wedge$ is taut. Hence $A^\wedge$ is weakly pre-adic by the equivalence of (2) and (1). Since the completion of a linearly topologized ring $A$ is complete (More on Algebra, Section 07E7) we see that $A^\wedge$ is weakly adic.

Let $A$ be weakly adic. Then $A$ is complete and and pre-admissible by (1) $\Rightarrow$ (3) and hence $A$ is admissible. Of course by definition $A$ has a countable fundamental system of open ideals. □

## References

[Gabber-Ramero] Gabber, Ofer and Ramero, Lorenzo. Foundations for almost ring theory–Release 7.5. arXiv preprint math/0409584, 2004.